DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

HERIOT-WATT UNIVERSITY

F78AA1 Actuarial and Financial Mathematics A

Date: 7 December 2012

Time: 09.30 to 11.30

Question	Marks
1	4
2	10
3	11
4	11
5	16
6	8
Total Marks	60

Attempt ALL SIX questions.

Approved electronic calculators may be used.

This version created on: November 28, 2012

1. Fill in the missing values in the following table, corresponding to an effective rate of interest of 7.5% per time unit:

i	v	d	$i^{(3)}$	$d^{(4)}$
0.075	?	?	?	?

[Total 4 marks]

2. (a) Define $A(s,t)$ to be the accumulation at time t of 1 invested at time s, and define $V(s,t)$ to be the present value at time s of 1 payable at time t (for $t \geq s$ in both cases). Assume that the following two conditions hold:

$$\begin{aligned} V(s,t) &= A(s,t)^{-1} \\ A(s,t) &= A(s,r)A(r,t) \qquad \text{for } s \leq r \leq t. \end{aligned}$$

Suppose cashflows $c_1, c_2, \ldots, c_n$ are payable as times $t_1, t_2, \ldots, t_n$ respectively. Let $ACC(t)$ be the accumulation at some time $t > t_n$ of these cashflows, and let $PV(s)$ be their present value at some time $s < t_1$. Prove that:

$$PV(s) = ACC(t) \times V(s,t).$$

[4 marks]

(b) The effective rate of interest will be 2% per year during 2013 and 2014, 0.5% per quarter-year thereafter. A series of cashflows is given as follows:

Date	Cashflow
1/1/13	+500
1/1/14	+300
1/1/15	−100
1/1/16	−200
1/1/17	−300

(i) What is the present value on 1 January 2013 of these cashflows?

(ii) What is the accumulation on 1 January 2018 of these cashflows?

[6 marks]

[Total 10 marks]

[PLEASE TURN OVER]

3. (a) Show that $(Is)^{i}_{\overline{n}|} = \dfrac{\ddot{s}_{\overline{n}|} - n}{i}$. [4 marks]

(b) A man buys an annuity designed to meet repayments under a special mortgage contract. The annuity is payable annually in arrears for 20 years. The first eleven payments are of £20,000 each. Thereafter, each payment decreases by £2,000 each year (so the 20th and final payment is of amount £2,000). The effective rate of interest will be 3% *per annum* during the first fifteen years, and 4% *per annum* during the last five years. Calculate the present value at outset of this annuity.

[7 marks]

[Total 11 marks]

4. You borrow £300,000 to buy a house. The loan has term 25 years, with level repayments payable annually in arrears. The interest rate when you take out the loan is 6% *per annum* effective.

(a) (i) What is the annual repayment?

(ii) Show that the capital outstanding just after the 10th repayment is £227,927.25. (Minor discrepancies due to rounding will not be penalised.)

(iii) What is the interest content of the 11th repayment?

[3 marks]

(b) Just after the 10th repayment has been made, you experience some financial difficulties and ask the lender if you may make no repayments in the next two years, then resume making level repayments annually in arrears for a further fifteen years (so the overall term of the loan is extended to 27 years). The lender agrees, but imposes the condition that the interest rate will increase to 9% *per annum* for the next **four** years, before reverting to its original level of 6% *per annum*. What is the new level annual repayment?

[8 marks]

[Total 11 marks]

[PLEASE TURN OVER]

5. You want to borrow £300 million to buy and operate a chain of hotels. You anticipate operating the hotels for 10 years, returning a profit of £55 million at the end of each year. At the end of the 10th year you will be able to sell the hotels for £80 million. You could raise the initial capital in one of two ways.

- A bank is willing to lend you the capital, charging interest at 8% *per annum* effective. You will use all net profits from the hotels to repay the loan as soon as possible. Once the loan is repaid, you will be able to invest the profits to earn 4% *per annum* effective.
- You could issue a bond of nominal value £300 million. It would be issued at par, would carry a coupon of 5% *per annum* payable yearly, and would be redeemed at 110% in ten years' time. You will be able to invest net profits to earn 4% per annum effective.

(a) Show that, on the basis of accumulated profit, issuing the bond is better than borrowing from the bank.

[10 marks]

(b) Suppose you do issue the bond. It is purchased by an investor who pays income tax at 25% and capital gains tax at 40%. What net redemption yield does the investor obtain on the bond?

[6 marks]

[Total 16 marks]

6. An investment fund has assets of £100 million on 1 January 2012. Investors pay additional amounts into the fund of £15 million on 1 July 2012 and £15 million on 1 January 2013, but withdraw £10 million from the fund on 1 July 2013. On 1 January 2014, the fund has assets worth £135 million.

(a) Suppose that you know that on 1 January 2013, just before the payment made on that date, the fund had assets worth £120 million. Find the linked internal rate of return on the fund, over the two calendar years 2012 and 2013.

[6 marks]

(b) Suppose that you also know that the fund had assets worth £100 million on 1 July 2012 and £130 million on 1 July 2013, in each case just before the payments or withdrawals made on those dates. What is the time-weighted rate of return on the fund during the years 2012–2013?

[2 marks]

[Total 8 marks]

[END OF PAPER]